

Multi-board Synchronization of n-channel Direct RF Digital Transceiver for Active Phased Array Radar using JESD204 Interface

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Abstract- Achieving synchronization in multi-channel transmit and receive systems has traditionally involved complex length-matched channel designs, posing challenges, particularly in higher-order channels. This paper introduces an innovative approach to multi-board synchronization of n-channel direct RF digital transceivers utilizing the high-speed serial JESD204 protocol, which leverages RF PLLs, high-speed serial DACs, and ADCs. The synchronization technique discussed herein significantly impacts digital beamforming (DBF) in active phased array RADAR systems. Instead of relying on a matched device clock or SYSREF signal, the proposed method employs phase delay calculation and PLL reprogramming to adjust phase differences among channels.

Synchronization is accomplished through online phase delay calculation within the FPGA using cross-correlation techniques between channels. Notably, this approach eliminates the need for a matched device clock or SYSREF signal, thereby reducing PCB complexity. Results underscore the method's efficacy, making it an efficient solution for active phased array RADAR systems, while a new method has been explored to analyze the impact of random phase changes on digital beamforming.

Keywords—RF DAC, RF ADC, SYSREF, JESD204, Digital Transceiver, Multi-channel Synchronization, Digital RF signal processing.

I. INTRODUCTION

Operation of Defence Radar/Communication equipment needs very high-power[8][9] systems to cover the military area. The design of energy-focused MIMO systems results in multi channels transmit and receive operation. The design & development of such a system is critical and challenging factors for almost all microwave and radio frequency equipment. To get the most efficiency in the multi-channel transceivers needs synchronization among the channels as well as the multi board. In the modern RADARs, communications, and instrumentation systems, to further increase the coverage, power, data rate, and trustability of the systems, system designers continue to place further emphasis on the multiple-antenna transmitter and receiver systems to continuously sampling the data from individual path RF analog-to-digital converters (RF-ADCs). Multiple-radiating element systems with digital beamforming (DBF) techniques also allow us to better focus on transmit power[1]. The system can reduce the size of a radiating element while increasing the transmitter coverage.

For similar multiple-antenna direct RF digital transceiver implementations, each individual transmitter requires modern RF digital-to-analog converters (RF-DACs) for digital data to RF transmission and RF-ADCs for RF-to-digital bits reception. Multiple receivers, transmitters, and the associated radiating element must also be time synchronized to achieve a coherent system[2][3]. The proposed design using RF-DAC and RF-ADC utilizes JESD204 subclass 1, which has the capability to achieve multiple device synchronization with some design considerations. The JESD204 interface solves this effectiveness problem by using multiple gigabit transceiver technology to interface between data converters (DAC and ADC) and an FPGA or ASIC[3].

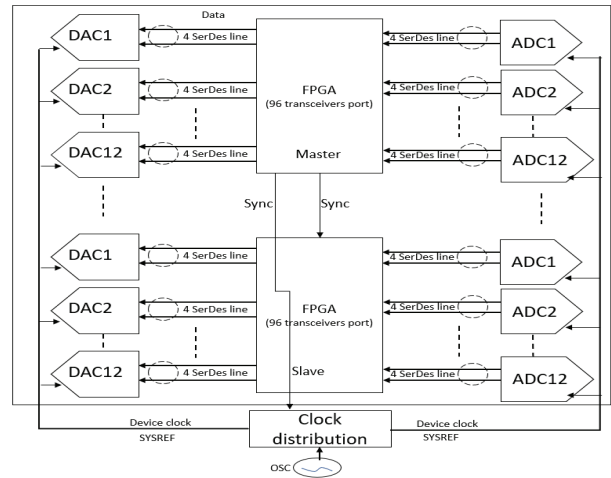


Fig. 1. Block diagram of n- channel direct RF digital transceivers

In the JESD204 system which is a serial protocol[4], the device clock and the system reference signal (SYSREF) play a critical role in ensuring all the data sampling devices and programmable logic devices in the system are synchronized to each other as shown in the figure 2.

For the synchronization of the JESD204-based RF-ADCs and RF-DACs, it is required all the clocks used in the system should be generated with the same phase lock loop (PLL) or with phase synchronized PLLs as shown in figure 3. One of the major challenging work in the n-channel digital transceiver is to achieve length matching across all the channels. The proposed innovation used the phase delay feature of PLLs to synchronize the clocks without having the length match or trace match paths [5].

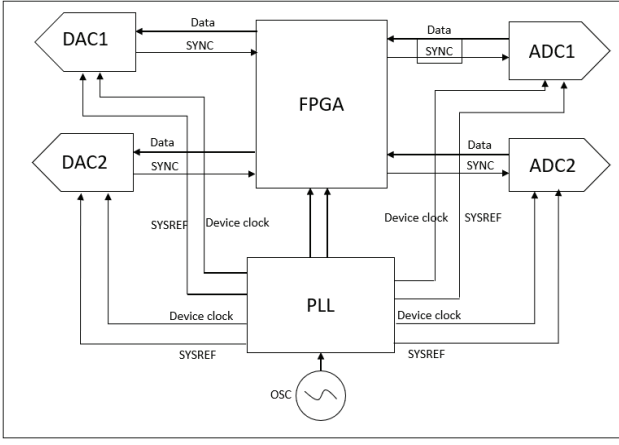


Fig. 2. Multi-device synchronization requirements in JESD204 systems.

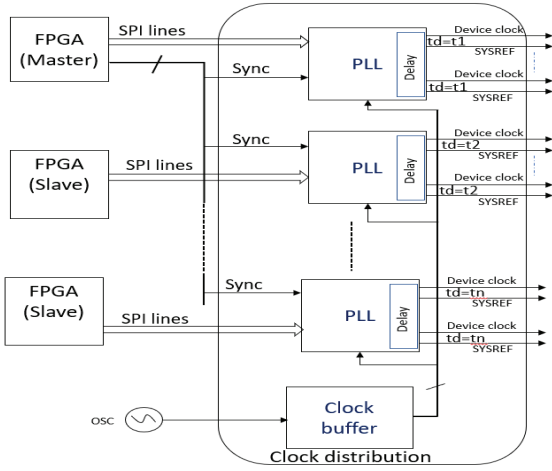


Fig. 3. Synchronization of multiple PLLs using FPGA (Master) sync

II. CONCEPTUAL DESIGN

A. SYSREF Timing Requirements and Frequency Selection for Multi Channels Synchronization

The SYSREF signal plays a crucial role in achieving system synchronization, demanding compliance with setup-and-hold times and a suitable frequency. High-speed devices, such as gigasample ADCs and RF-sampling DACs, pose challenges in SYSREF setup-and-hold timing, potentially necessitating dynamic delay adjustments or external calibration procedures. JESD204B offers flexibility in data converter clocking, particularly with integrated PLLs that facilitate lower-frequency device clocks, simplifying setup-and-hold requirements.

Choosing the SYSREF frequency is constrained by the requirement that it operates at a frequency equal to or at an integer division of the local multi-frame clock (LMFC) frequency. SYSREF is calculated using Equation 1, where Lane Rate is the interface bit rate of the serializer/deserializer (SerDes), F is the number of multi-frames in a frame (typically 1 or 2), K is the number of frames per multiframe block, and N is a positive integer (usually 1, 2, or 4).

$$f_{\text{SYSREF}} = \frac{\text{Lane Rate}}{10 * F * K * N} \quad (1)$$

$$\text{Lane Rate} = \left(\frac{M}{L}\right) \times NP \times \left(\frac{10}{8}\right) \times \text{Data rate} \quad (2)$$

The data rate is the rate at which data is sent to the JRx from the JT_x, in samples per second (sps), L is the JESD204 parameter for lanes per link, M is the JESD204B parameter for converters per link and NP is the JESD204B parameter for the total number of bits per sample.

Constraints may also arise if SYSREF is involved in the synchronization of other digital features, impacting its frequency alignment with both the LMFC frequency and the lowest internally-generated frequency. Adjustments may be necessary to meet these constraints.

B. RF PLLs Synchronization[5]

For the synchronization of the n-channel digital transceiver, multiple RF PLLs are used as shown in figure 3. All PLLs are synchronized using the rising edge of the sync signal which is generated from the master FPGA. In the proposed innovation master and slave FPGA configuration is proposed. For the synchronization of PLLs, it is mandatory that all the sync signals should be generated with a common source. For the configuration of PLLs using SPI protocol, any FPGA (master or slave) can be used depending on the FPGA capacity and input/output (IO) pins. For the programming of PLLs, the following steps are used.

Step 1: Provide the common reference clock using clock buffers to all the PLLs.

Step 2: Write the configuration registers of all the PLLs at a time or one by one.

Step 3: Wait for the lock signal of all PLLs

Step 4: When all PLLs are locked, trigger the sync signal high a few times (depending on the PLL recommendation in the datasheet) and then make it lower permanently.

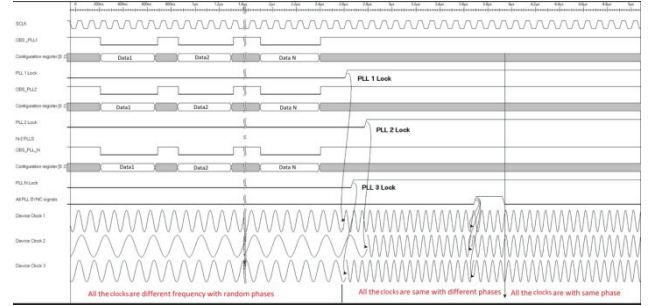


Fig. 4. Timing diagram of multi-PLLs synchronization scheme using sync signal

The timing diagram illustrating the synchronization scheme for multiple PLLs employing a sync signal is depicted in Figure 4. Initially, all PLLs undergo configuration through SPI, and their device clocks operate with varying frequencies and phases due to being in an unlocked state. Following the configuration register writes, the PLLs are individually locked at different time instances to achieve the same frequency, albeit with potential phase differences.

A sync trigger is generated by the Master FPGA to reset the counters of all PLLs. Consequently, their outputs become synchronized with each other, resulting in all PLLs sharing the same frequency and phases, as illustrated in Figure 4.

Through these procedures, all PLLs are brought into synchronization with the same phase. The phase of each individual clock can be adjusted based on the positions of the

ADCs and DACs on the board. Additionally, the PLLs used in this setup allow for the modification of path delay through the configuration register

C. Path delay or phase difference calculation using a cross-correlation technique

The cross-correlation technique is used to calculate the phase difference between the channels.

MATLAB code for delay calculation and correction

```
[c,lags]= xcorr(s1,s2);
[~, iLag] = max(c(find(lags==0): end));
s4=circshift(s2,[0 iLag]);
phase_shift_cal=2*pi*(iLag-1) *fins;
```

Where fins= fin/fs, fin = input frequency and fs = sampling frequency

After calculating the phase difference, the path delay between the channels is calculated using the below equation.

```
ang_deg=phase_shift_cal*180/pi
td=ang_deg/360/fin
```

td is the time delay in second

The above process is repeated for all the channels and PLLs are reprogrammed to adjust the delays. As the above process is implemented in real-time, it makes it a fully automatic process for the synchronization of the n-channel digital transceiver within a few seconds.

D. Digital beamforming for the validation of random phase change in the receive signal

The impact of phase change in the receive with due to un-synchronization or clock jitter, digital beam forming for the 32 channels of receive signal has been carried out. In digital beamforming for a receiver end, the basic concept involves combining signals from multiple antennas with different weights and possibly different time delays to maximize the signal strength from a desired direction or to minimize interference from other directions. The specific formula for digital beamforming can depend on the algorithm used. A common approach is the Delay-and-Sum beamforming algorithm [7].

For a simple Delay-and-Sum beamformer at the receiver end, the combined signal ($S_{\text{beamformed}}$) can be expressed as a weighted sum of the received signals ($x_i(t)$) from individual antennas:

$$S_{\text{beamformed}}(t) = \sum_{i=1}^N w_i \cdot x_i(t - \tau_i) \quad (3)$$

$S_{\text{beamformed}}(t)$ is the combined (beamformed) signal at time t at the receiver. N is the number of antennas in the array. w_i is the complex weight applied to the signal from the i -th antenna. $x_i(t)$ is the signal received by the i -th antenna. τ_i is the time delay applied to the i -th antenna's signal to steer the beam in the desired direction. The weights w_i and time delays τ_i are determined based on the desired beamforming characteristics, such as directionality and beamwidth.

III. IMPLEMENTATION

The implementation process involved several key steps and considerations to ensure its successful execution

A. Hardware Setup and PCB Design:

This involved selecting the appropriate components, such as RF PLLs, high-speed serial DACs, and ADCs. RF PLLs are available in which we can program the delay of individual channels.

In this demonstration, I have selected the clock buffer from the analog devices [6] which can provide the programmable delays in channels. We can use the following delays

TABLE I. ANALOG FINE DELAY:

Adjustment Range	:135 to 670 ps
Resolution	:25 ps
Maximum Analog Fine Delay Frequency	:3200 MHz

TABLE II. COARSE DELAY:

Adjustment Range	:0 to 17 (17 delay steps in $\frac{1}{2}$ VCO period)
Resolution	:169.54 ps
Maximum Analog Fine Delay Frequency	:3200 MHz

Total delay (T_{total}) which can be tuned is 3.326 ns.

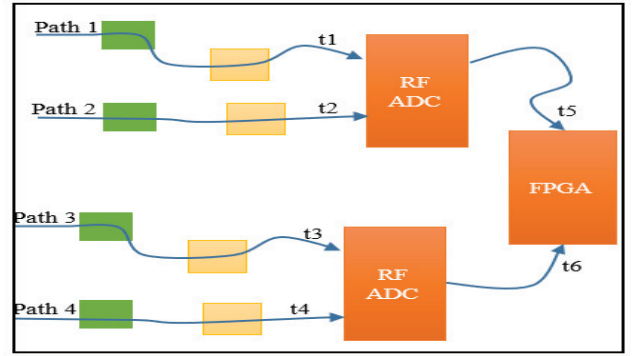


Fig. 5. Approximate PCB signal routing with path delay

If the difference from the maximum path delay ($t1+t5$) to minimum path delay ($t4+t6$) is less, then total then by using the above RF PLL module synchronization is easily achieved and PCB complexity reduces. To reduce PCB complexity, we carefully designed the layout to accommodate the multi-board setup. Minimizing signal interference and ensuring proper signal integrity were the primary concerns during the PCB design phase.

B. Synchronization Methodology:

The synchronization methodology was a crucial aspect of the implementation. We opted for a novel approach that utilized phase delay calculation and reprogramming of the RF PLLs to adjust the phase difference between the channels. By calculating the phase delay online in the FPGA using the cross-correlation technique, we could achieve precise synchronization without relying on a match-matched device clock or SYSREF signal.

C. Testing and Validation Setup: Extensive testing and validation were conducted at various stages of the implementation process. We performed thorough testing to verify the synchronization performance, measure phase differences, and evaluate the overall efficiency of the system. Any issues or discrepancies encountered during testing were addressed promptly to refine the implementation.

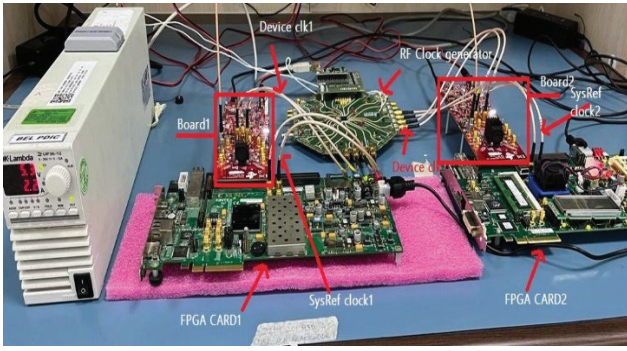


Fig. 6. Proposed Test setup for 2 board synchronization with common RF PLL for clock generation

Board1 and Board2 of the RF transceiver are connected with 2 different FPGA boards. Both the RF transceivers are given the clock through a common RF PLL with unmatched cable. RF PLL is programmed with PC which is configured as a master and both the RF transceiver are configured with same FPGA1 and FPGA2. After calculation of the path difference between the two path of RF clock, RF PLL has been reprogrammed with individual delay. Hence the synchronization is achieved in ns.

Overall, the implementation phase was a challenging yet rewarding process. With the successful deployment of the new technology project, we achieved efficient synchronization among the n-channel direct RF, digital transceiver units using the high-speed serial JESD204 protocol.

IV. RESULTS

The methods mentioned above have been executed and evaluated based on the test setup outlined in Figure 6. Initially, synchronization of two RF PLLs is accomplished following the timing diagram depicted in Figure 4. Both PLLs undergo configuration via SPI, and after achieving lock, a sync trigger is applied. Subsequently, the FPGA captures the outputs of both PLL clocks, revealing that both clocks share the same frequency and phase, as illustrated in Figure 7.

A test is conducted involving the generation and reception of an L-Band frequency, illustrated in Figures 8 to 11. However, it's important to note that the method proposed above is not constrained to a specific frequency band. In this test, the L-Band signal is received through multiple RF ADCs and digitally downconverted to an intermediate frequency (IF), as indicated in Figure 8. The path delay among the three channels is calculated, considering the first channel as the reference. Subsequently, PLLs are reprogrammed to achieve synchronization. After successful synchronization, data from all three channels are plotted in MATLAB, as depicted in Figure 9.

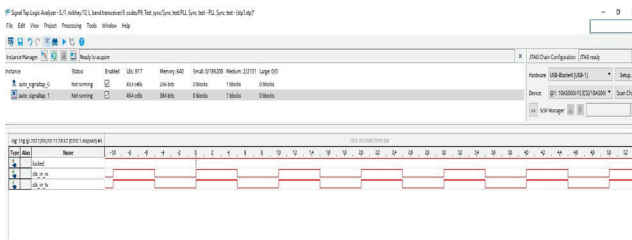


Fig. 7. the FPGA implementation of two PLLs, namely clk_in_rx and clk_in_tx, displaying identical phases for their respective clocks

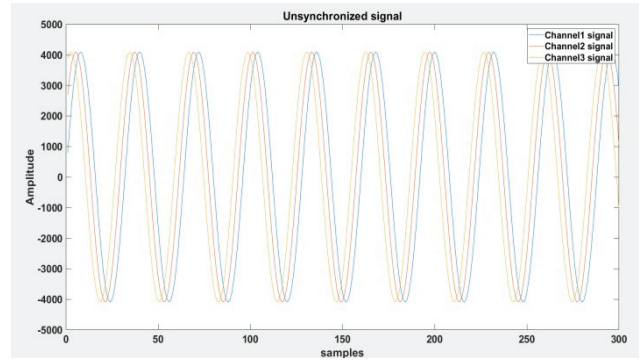


Fig. 8. Unsynchronized received L-Band signal using RF ADC

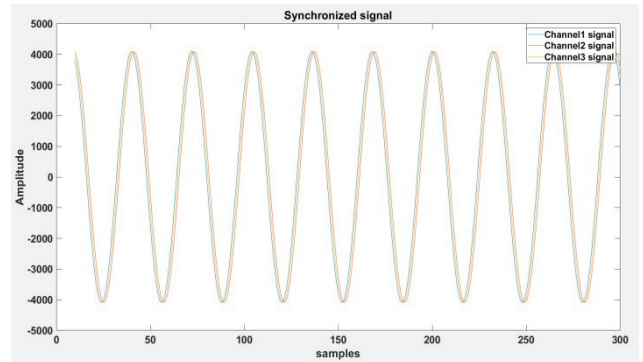


Fig. 9. Received L-band signal after synchronization using RF ADC

In the same way, all the steps are repeated for the transmit signals, and before and after synchronization results of transmit channels are shown in figure 10 and figure 11 respectability.



Fig. 10. Transmitted unsynchronized signals using RF DAC

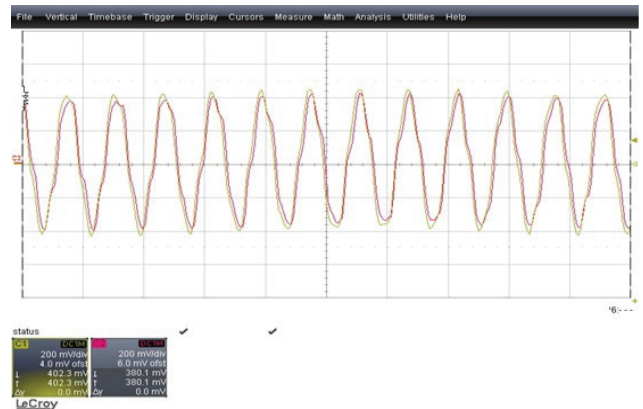


Fig. 11. Transmitted synchronized signals using RF DAC

The study investigates the repercussions of random phase variations resulting from the lack of synchronization in receive channels. In Figure 12, it is observed that when all 32 channels are synchronized, forming the same phase, a main beam emerges at 0 degrees of elevation with a gain of 30 dB.

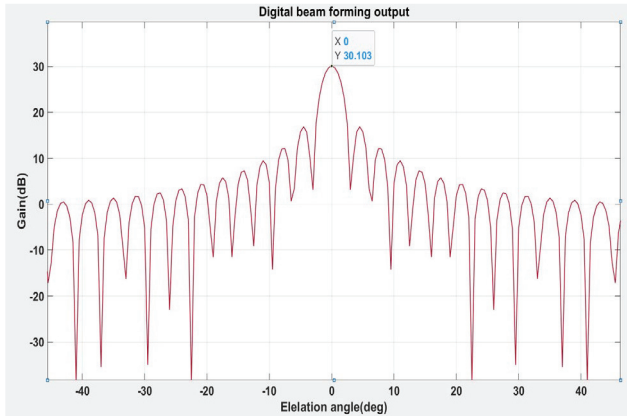


Fig. 12. Beam forming out with 32 Rx channel having same synchronization in the channels

To analyze the impact of phase alterations in the beamforming technique, MATLAB is utilized with Equation 3. The phase is treated as randomly ranging from -10 degrees to 10 degrees. To account for phase randomness, the experiment is conducted 14 times. The elevation angle of the main beam undergoes random shifts between -10 degrees and 10 degrees, as depicted in Figure 13.

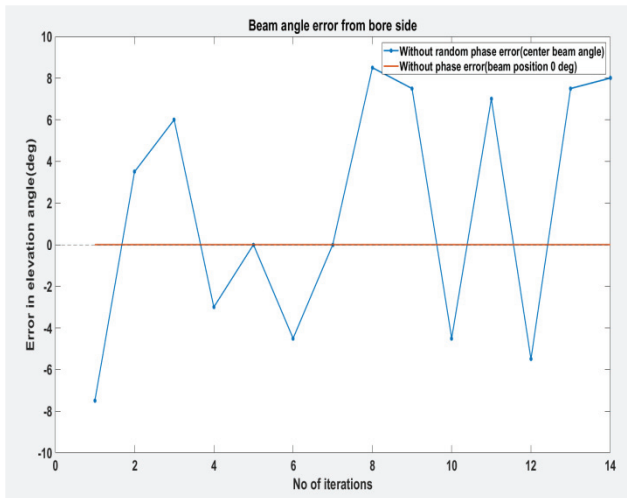


Fig. 13. Error in beam elevation angle with change in the phase of -10 deg to 10 deg.

V. CONCLUSION

In conclusion, this research paper introduces a pioneering and efficient approach to achieving multi-board synchronization for n-channel direct RF digital transceivers using the high-speed serial JESD204 interface. The application of this method holds promise for advancing the performance and capabilities of defense radar and communication systems, particularly those employing active phase array radar technology.

The paper effectively addresses the substantial challenges associated with attaining precise synchronization across multiple channels and boards. Through the strategic use of RF PLLs, high-speed serial DACs, and ADCs, the proposed method demonstrates efficient synchronization without

relying on matched device clocks or SYSREF signals. The conceptual framework of the synchronization method is grounded in the coordination of multiple RF PLLs using FPGA, ensuring the generation of clocks with identical phases. Crucially, real-time accurate synchronization between channels is achieved through the incorporation of phase delay calculation and cross-correlation techniques.

The presented results, including the successful generation and reception of L-Band and S-Band signals, validate the efficacy and versatility of the proposed method. The achieved synchronization accuracy within nanoseconds underscores its potential for cutting-edge applications in defense radar and communication systems. Furthermore, the study investigates the impact of random phase changes in receiver signals from different antenna elements on the main beam of beamforming, providing valuable insights into the method's robustness and performance under varying conditions.

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